

Linear Mixed Models Project

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Study objectives and context

- Study realized in Belgium with 1000 overweight students
- **Objective:** Evaluate the effect of three prevention approaches on BMI evolution over one academic year, while accounting for baseline characteristics (gender, smoking, physical activity).
- **Approach 1:** Information booklet on healthy lifestyle and overweight prevention.
- **Approach 2:** Information sessions, booklets, and sports activity discounts.
- **Approach 3:** Approach 2 + health coaching app with tracking and guidance.

Body Mass Index (BMI)



Research question and data structure

Research question: *How does BMI change over time under each prevention approach, after adjusting for baseline characteristics?*

Longitudinal study:

- repeated BMI measurements on the same students (T1–T5)
- Repeated observations from the same student are correlated
- Therefore, a mixed-effects model is recommended

Variables

- **ID:** Student identifier
- **Baseline covariates:** Gender, Smoking, Physical activity
- **Group:** Prevention approach (1–3)
- **Outcome:** BMI measured at Baseline and T1–T5

Wide format							
Original dataset							
ID	Group	BMI _b	BMI _{T1}	BMI _{T2}	BMI _{T3}	BMI _{T4}	BMI _{T5}
1	3	30.2	32.3	32	29.7	28.4	30.3

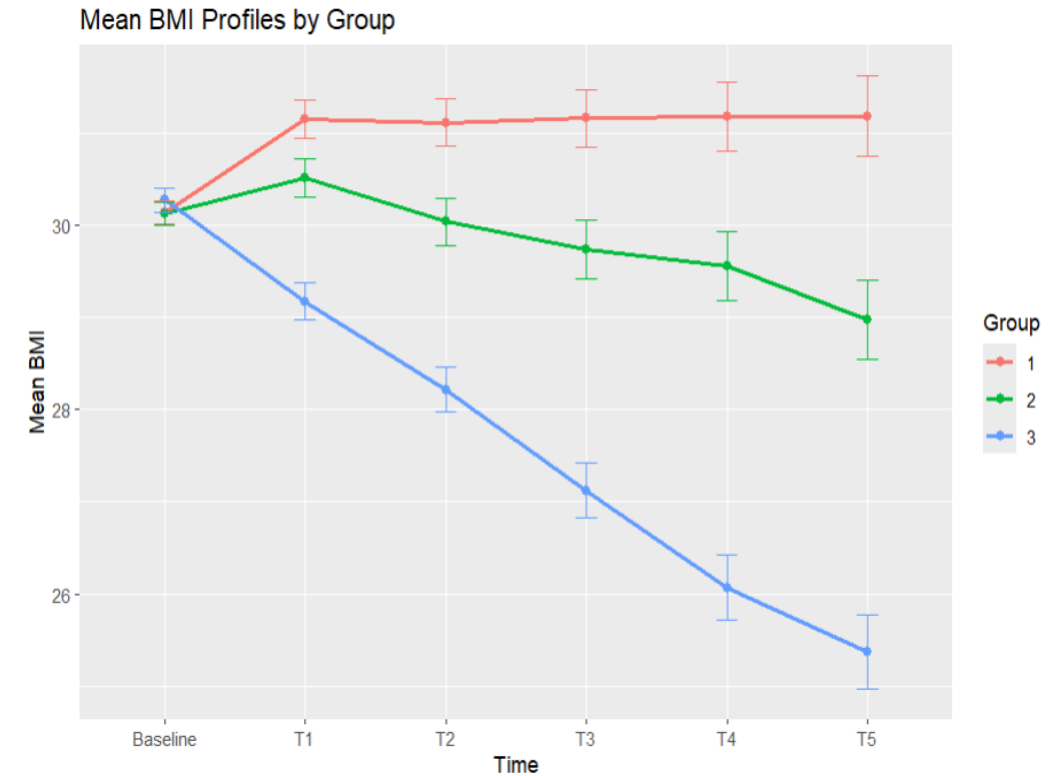


Long format			
Used for analysis			
ID	Group	Time	BMI
1	3	Baseline	30.2
1	3	T1	32.3
1	3	T2	32.0

Exploratory analysis

Mean BMI trajectories by intervention group

- **Group 1:** slight initial increase followed by stabilization.
- **Group 2:** gradual decrease after T1.
- **Group 3:** marked and nearly linear decrease throughout follow-up.
- These descriptive trends suggest that BMI evolution may differ between intervention groups



Inference methodology

Step 1: Start with the fullest model that addresses the question

Fixed effects: Time*Group, Smoke, LIVE, Gender

Random effects: subject-specific random intercept, slope for time

Step 2: Choose the random-effects structure

- Compare with LRT: random intercept **vs** random intercept + random slope

Step 3: Selection of residual covariance structure

- Compare heterogeneous variances and CS structures via AIC

Step 4: Fixed effects selection

Test whether Group x Time, Gender, Smoke, LIVE are significant

Step 5: Diagnostics

On the selected model: Residuals vs fitted graph, QQ-plot of residuals and random effects

Choice of random effects structure

m1: $BMI_{ij} = \beta_0 + \beta_1 Time_{ij} + \beta_2 Group_j + \beta_3 (Time \times Group)_{ij} + \beta_4 Gender_j + \beta_5 Smoke_j + \beta_6 Live_j + b_{0j} + b_{1j} Time_{ij} + \varepsilon_{ij}$

- **#Parameters: 26** = 1 + 5 + 2 + 10 + 1 + 1 + 2 + 1 + 1 + 1
- i- time, j- patient
- *Time, group, Smoke, Live, Gender* are treated as categorical fixed effects factors
- **Interpretation m1:** Students differ both in baseline BMI and in their rate of BMI change over time.

m2: Same fixed effects, but $b_{1j} = 0$

Interpretation m2: Students have different baseline BMI values but are assumed to follow the same average trajectory over time.

LRT test	anova(m2,m1)	P-value: <0.0001
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Result: Random slopes did not significantly improved model fit, indicating that students differ in baseline BMI but not in their BMI evolution over time.

Retained: m1 → random intercept model+ random slope model

Selection of residual covariance structure

Model	df	Residual variance	Residual correlation	Assumption	AIC
<i>m1</i>	26 22+3+1	Homogeneous	None	Same residual variance at every visit	25977
<i>m3</i>	31 22+3+1+5	Heterogenous	None	Residual variance differs across visits	25696
<i>m4</i>	27 22+3+1+1	Homogeneous	CS	Constant residual correlation within students	25979

Model limitation: Complex residual correlation structures (e.g., AR(1) and UN) could not be reliably fitted due to convergence issues and were not retained.

Selected structure: *m3* — heterogeneous residual variances by measurement occasion

- Residual variability changes across measurement occasions

Fixed effects selection

- ML is used to compare models with different fixed-effects structures.

Fixed effect	Time	Group	GENDER	SMOKE	LIVE	Time x Group
df	5	2	1	1	2	10
F	25.49	4.13	68.30	0.04	1.47	14.56
p-value	<0.0001	0.0163	<0.0001	0.838	0.231	<0.0001

1. **Time × GROUP:** BMI trajectories differed significantly between groups.

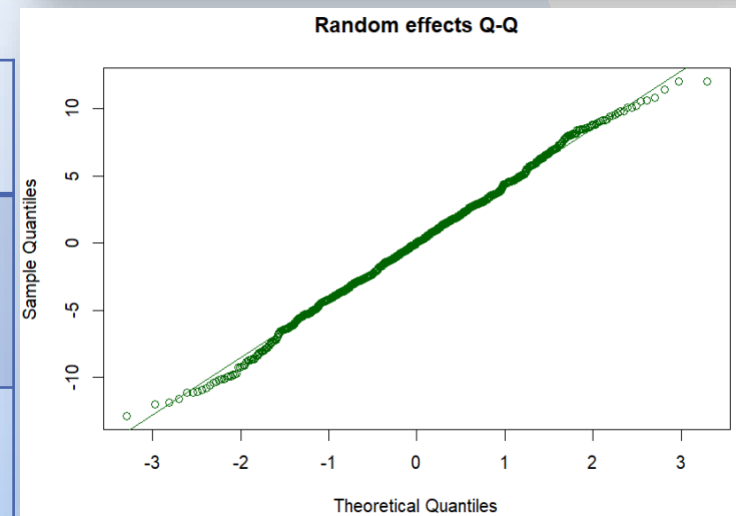
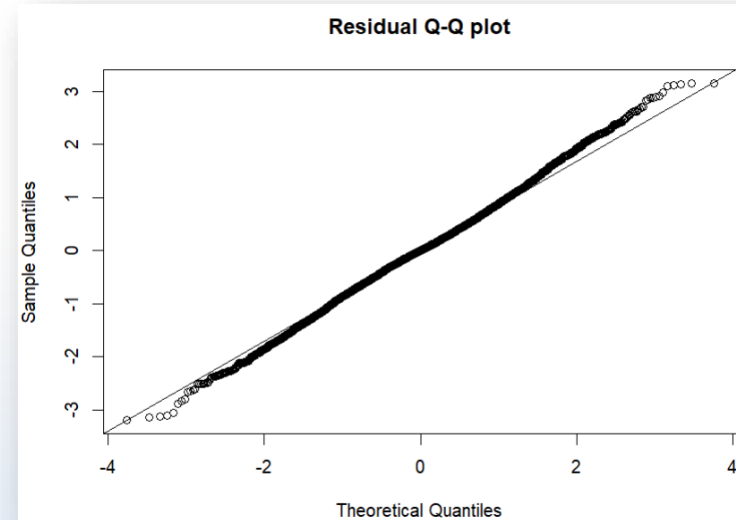
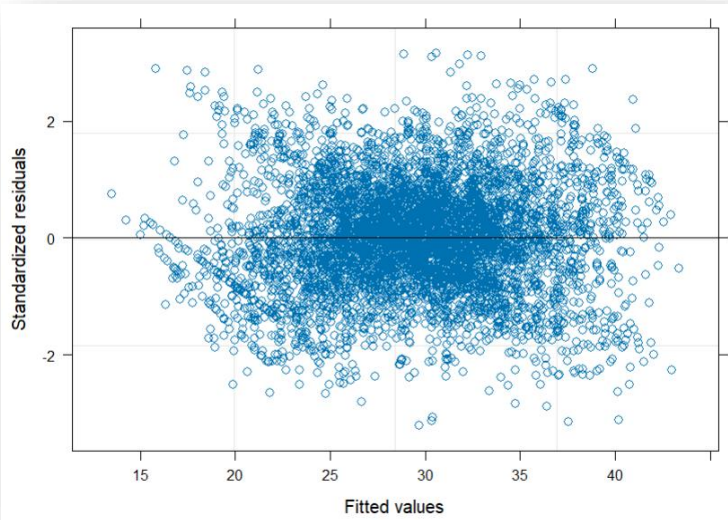
2. **Gender:** BMI trajectories differed significantly between genders

Joint LRT test	Models compared	Δdf	LRT statistic	p-value	Conclusion
SMOKE + LIVE	With vs. without SMOKE & LIVE	3	4.34	0.397	The joint effect not statistically significant

Model diagnostics

Final model:

1. $BMI \sim Time \times GROUP + GENDER$
2. Random intercept and random slope for each student
3. Heterogeneous residual variance across Time



Diagnostic	What to check	Conclusion
Residuals vs fitted	Linearity / systematic patterns	✓ No major pattern
Residual Q-Q plot	Normality of residuals	✓ Approximately normal
Random-effects Q-Q plot	Normality of random effects	✓ Approximately normal

Overall: No major violations of the LMM assumptions were detected.

Conclusion

Main finding:

- **Time × GROUP interaction was significant**

→ BMI trajectories differed significantly between the three prevention approaches. (slide 8)

Intervention effect

- **Approach 1:** Adjusted BMI increased by approximately **1.1 BMI units** from baseline to T5.
- **Approach 2:** BMI progressively decreased relative to Approach 1.
- **Approach 3:** Showed the **largest BMI reduction**
- At T5, BMI was **3.64 units lower than Approach 2** (Tukey-adjusted $p < 0.0001$)

Other findings

- Groups were comparable at baseline
- Gender was significantly associated with BMI. (slide 8)
- Residual variability differed by Time → heterogeneous variance supported. (slide 7)

- **Overall, Approach 3 appears to be the most effective prevention strategy for reducing BMI over the academic year.**

Approach 3:
Comprehensive
lifestyle support
with coaching,
tracking &
guidance